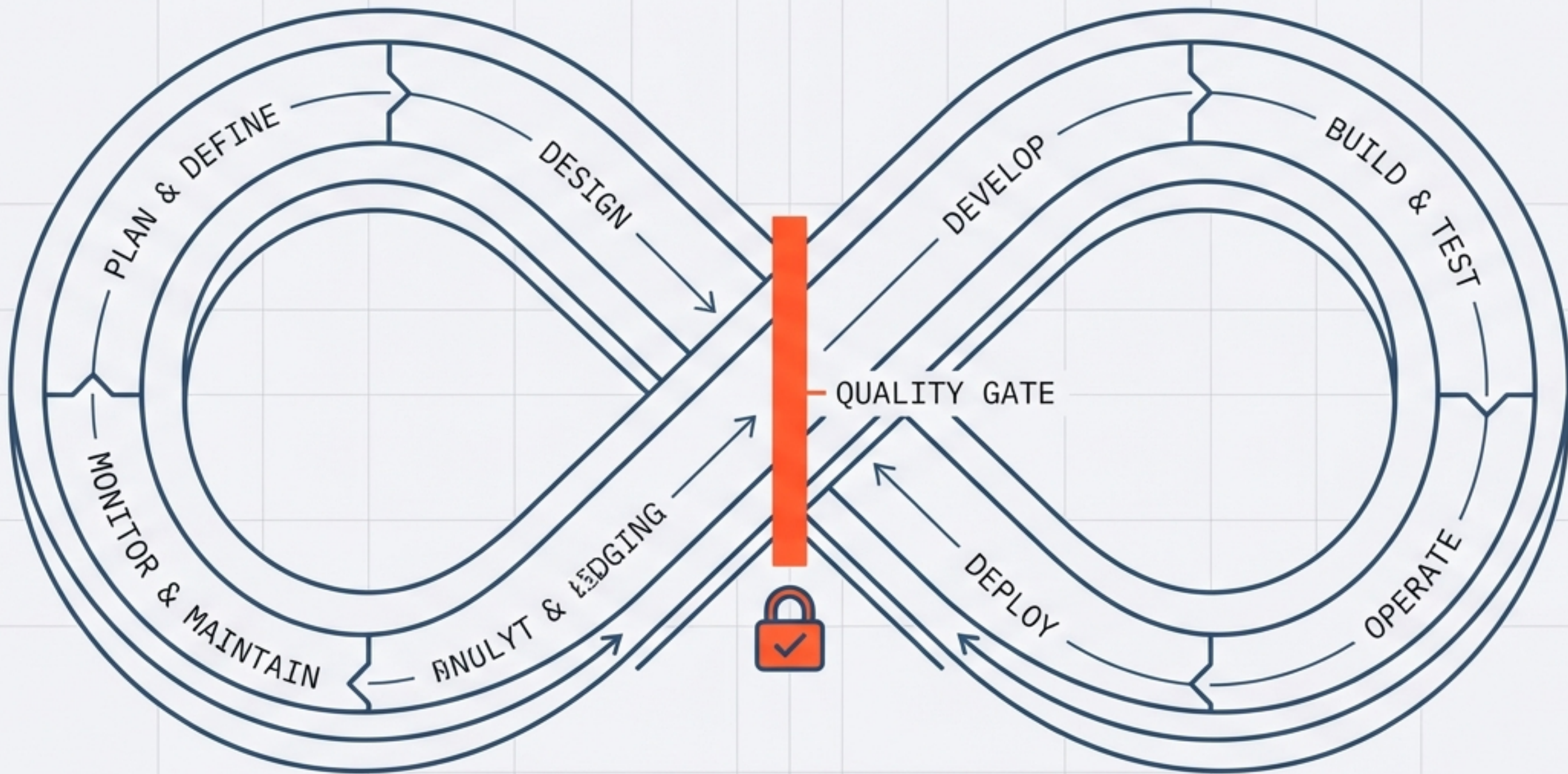
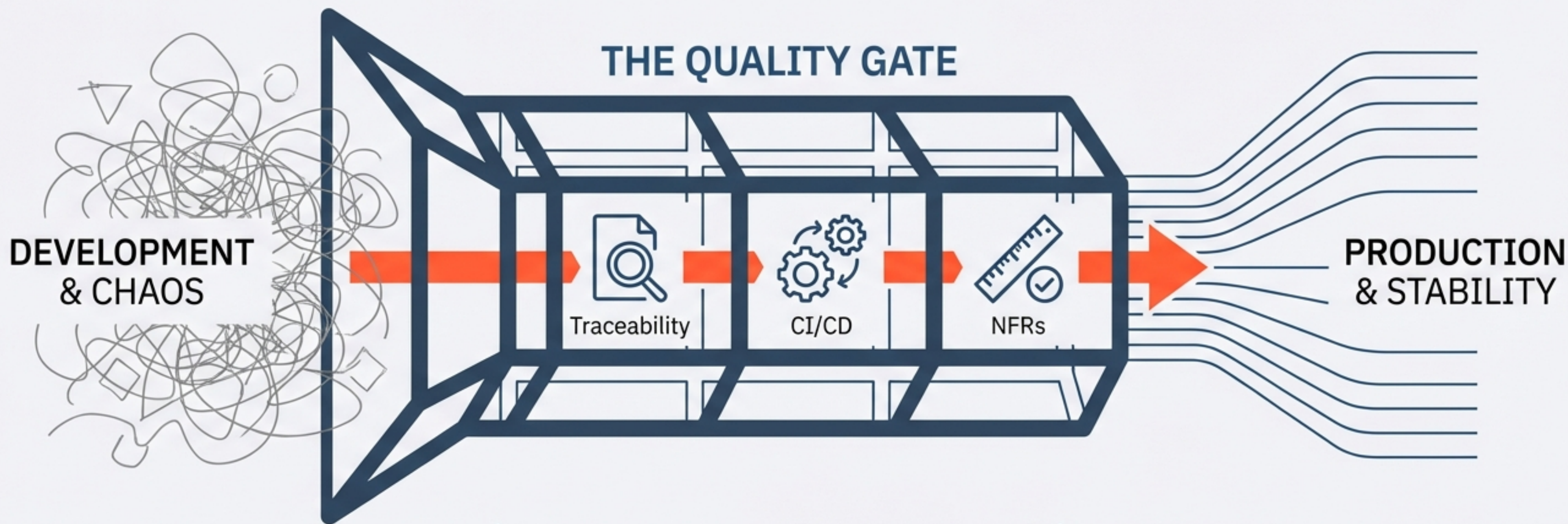


Software Testing, Deployment, & Maintenance Strategies

A Systems Analyst's Framework for Production Readiness



Quality is a Managed Habit, Not an Accident



The Mandate

Poor testing leads to failure.
The goal is production readiness.

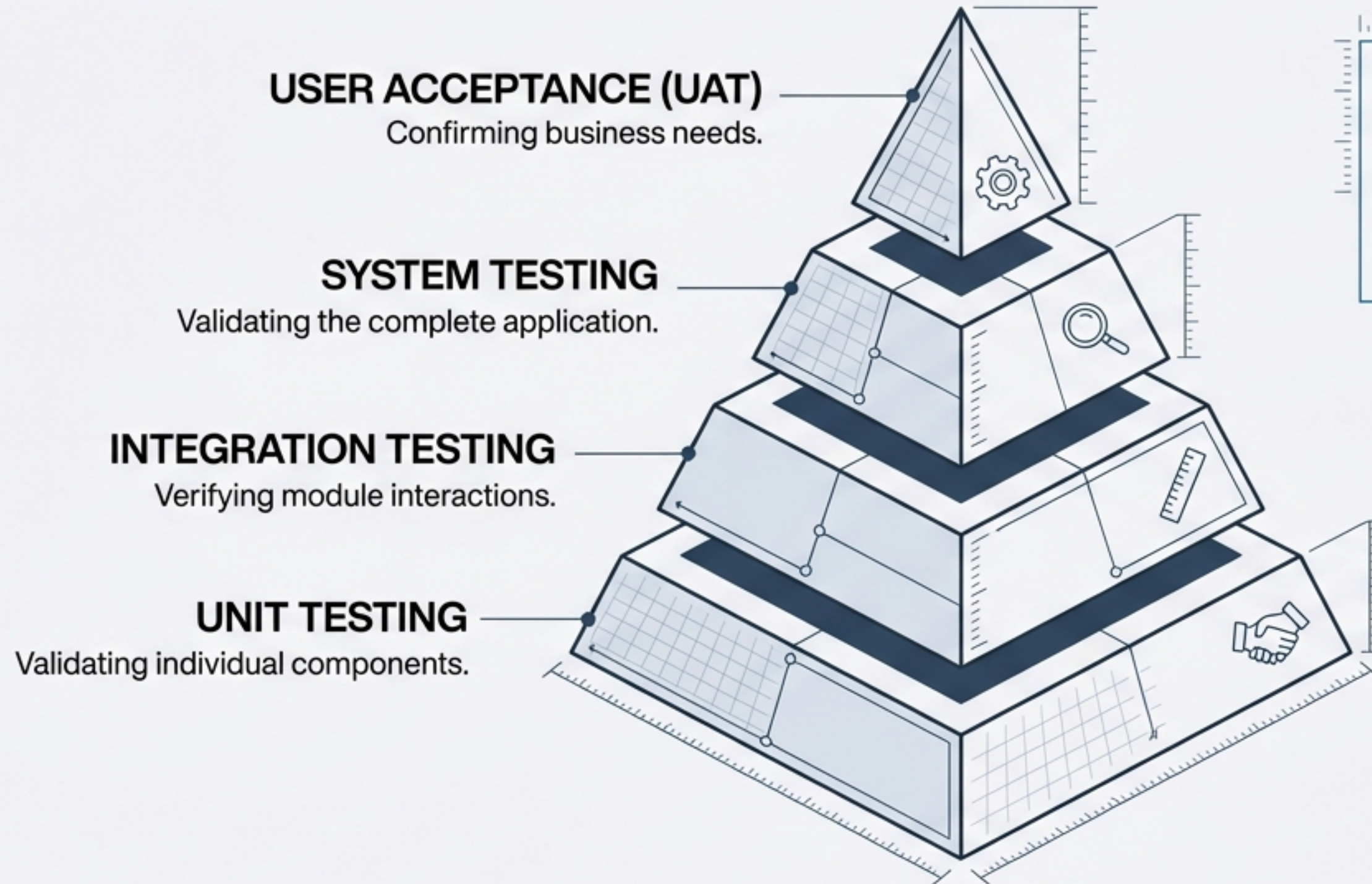
The Tools

- Traceability
- CI/CD
- Non-Functional Requirements

Analyst Role

Bridging business needs and technical execution.

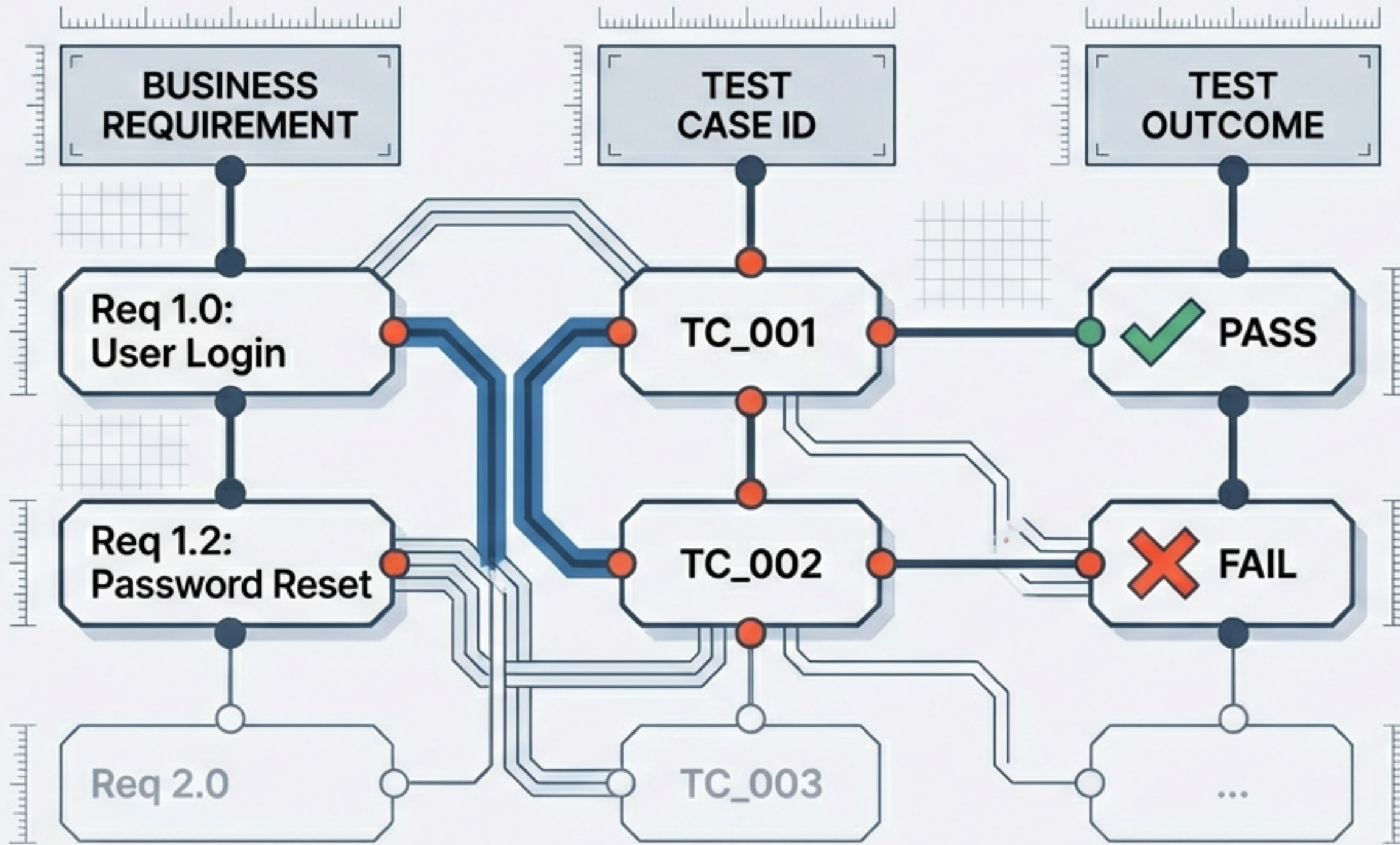
The Hierarchy of Defense: Testing Levels



THE ANALYST LENS:

Analysts contribute at every stage, ensuring technical tests align with business logic.

Ensuring Coverage with Traceability Matrices

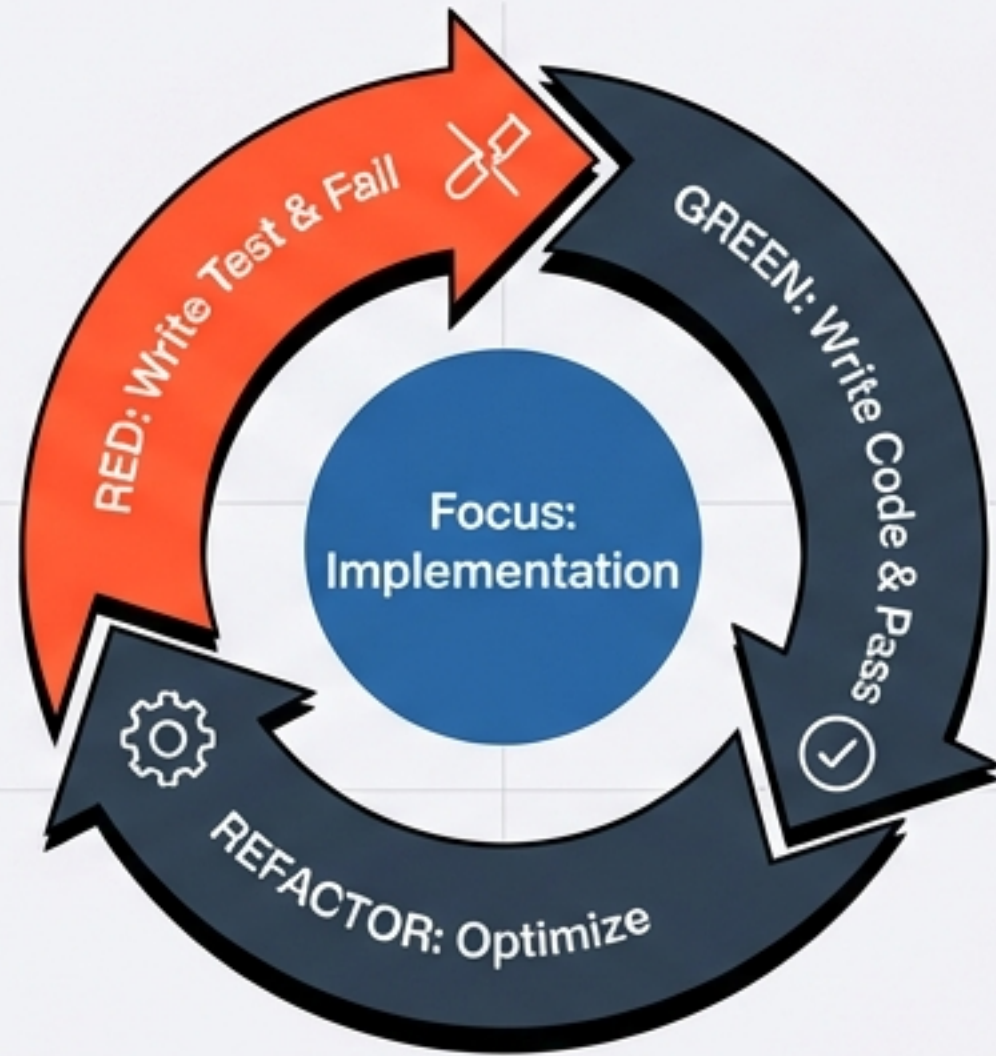


Benefits of the Matrix:

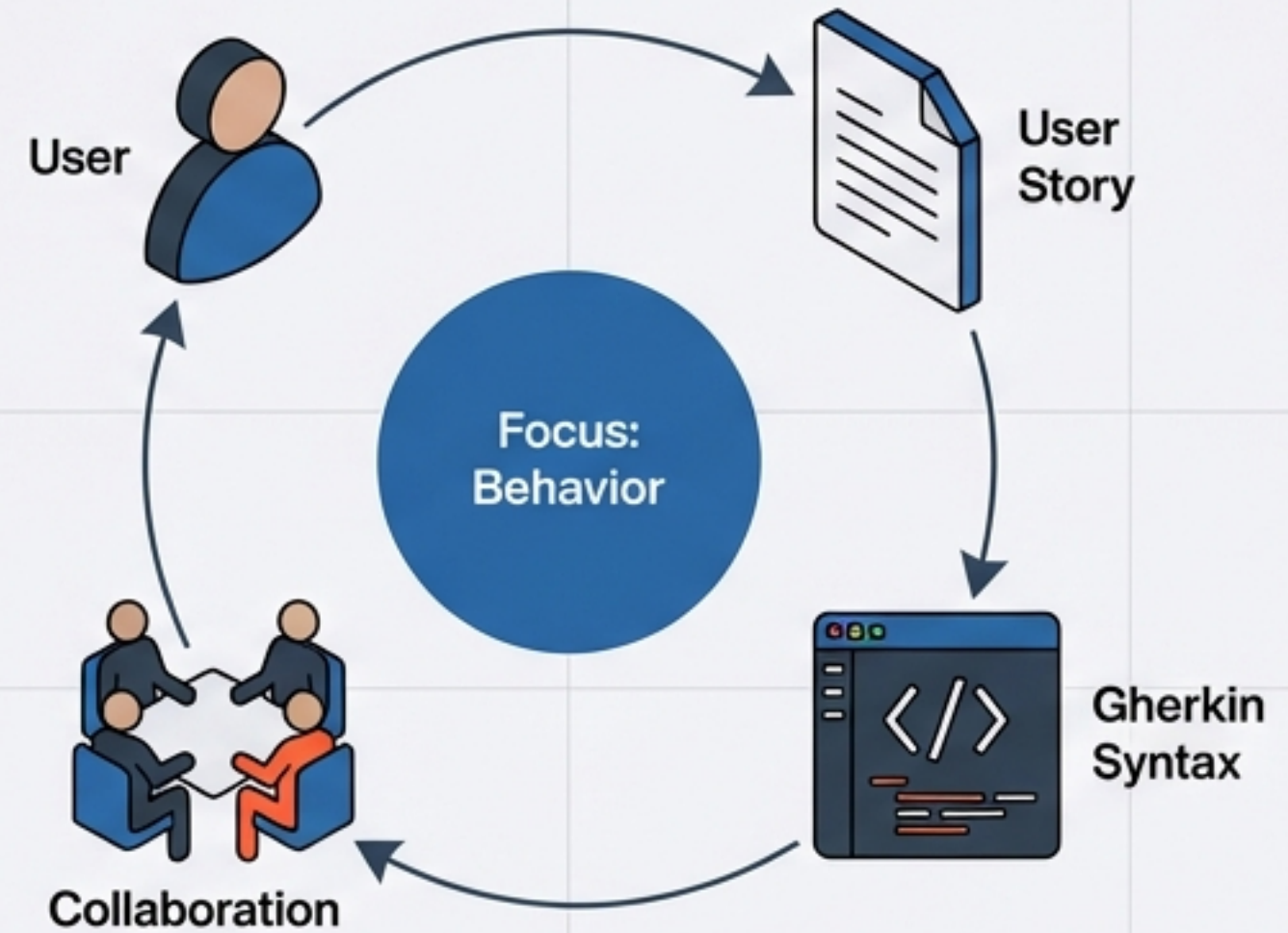
- Prevents Scope Creep
- Ensures 100% Test Coverage
- Coordinates QA, Dev, and Stakeholders

Shifting Left: TDD & BDD Methodologies

TDD (Test Driven Development)



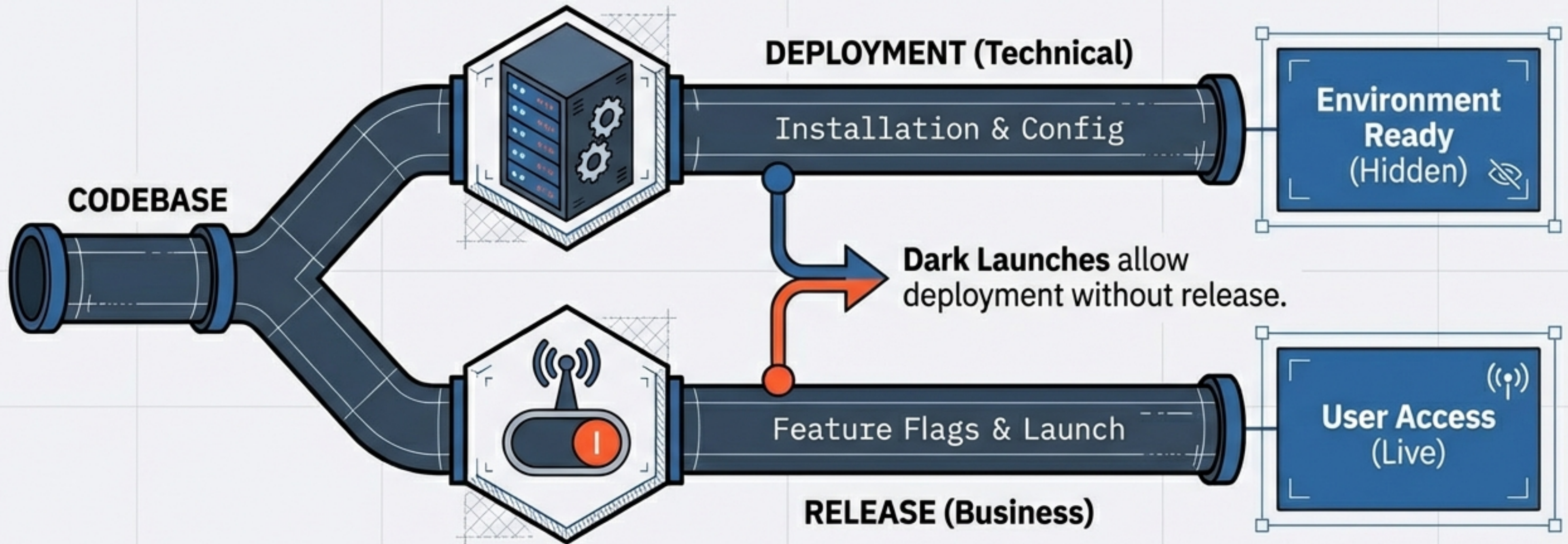
BDD (Behavior Driven Development)



Gherkin Syntax Example:

Given [Context], When [Action], Then [Outcome].

Deployment ≠ Release

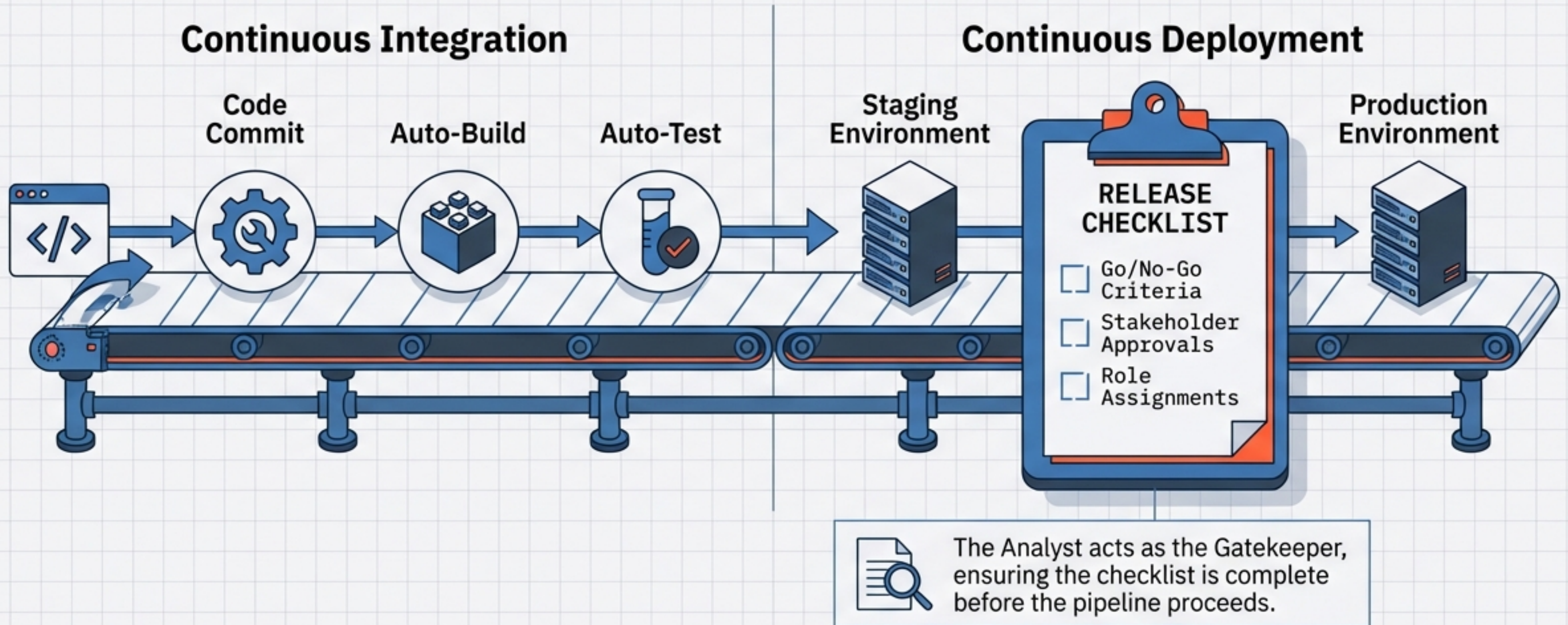


Analyst Note:

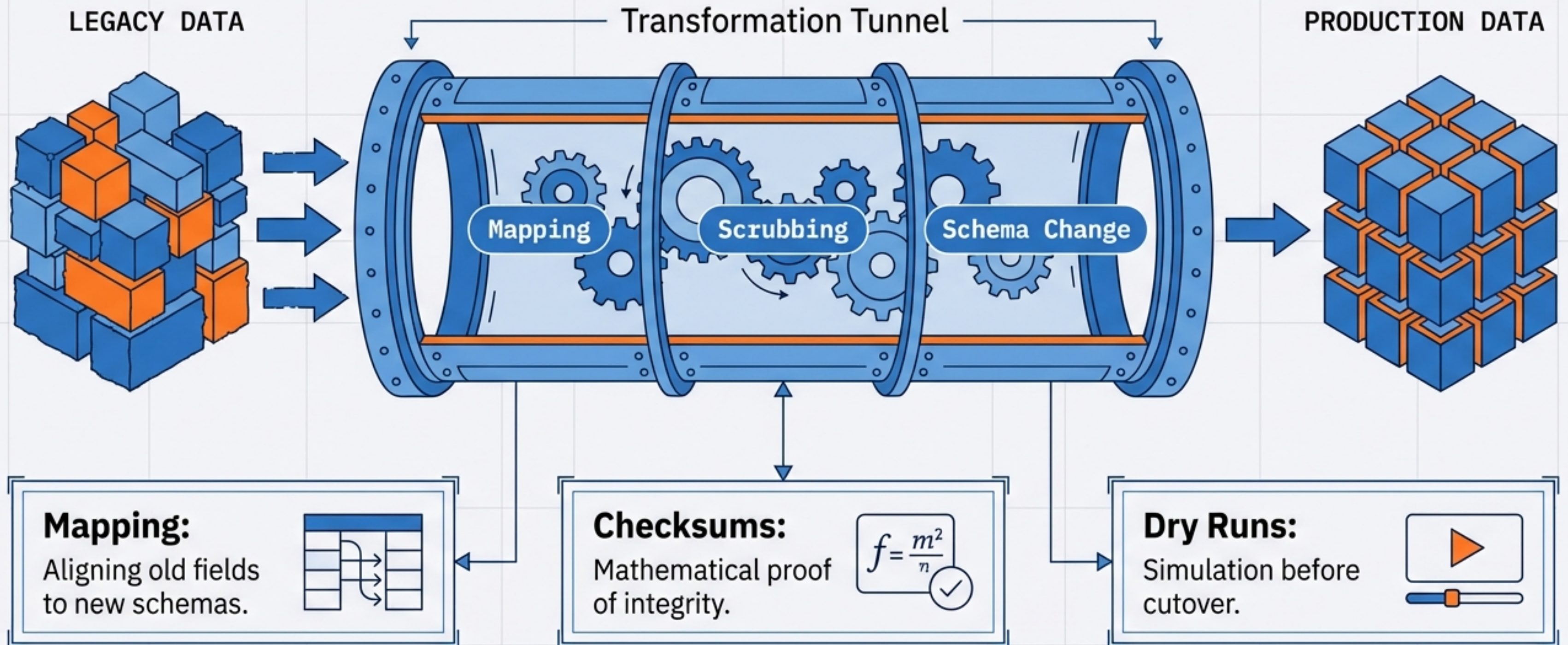
The Analyst ensures technical deployment supports the business release strategy.

The Delivery Pipeline: CI/CD & The Release Checklist

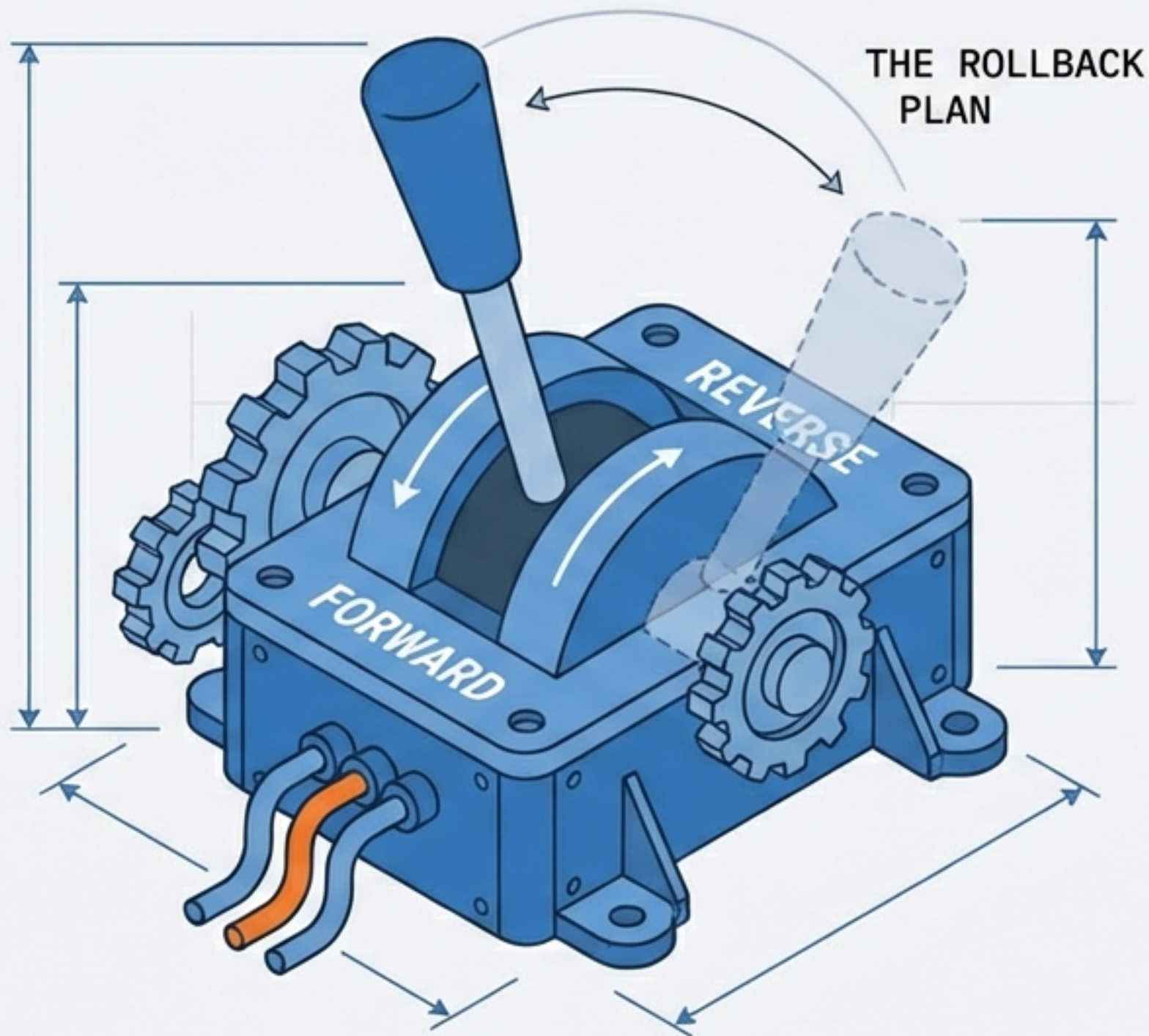
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Data Migration: Mapping Legacy to Future



Risk Management & Rollback Strategies



Safety Net Strategies



Blue/Green Deployment
Traffic switches between two identical environments.



Canary Releases
Roll out to a small % of users first.

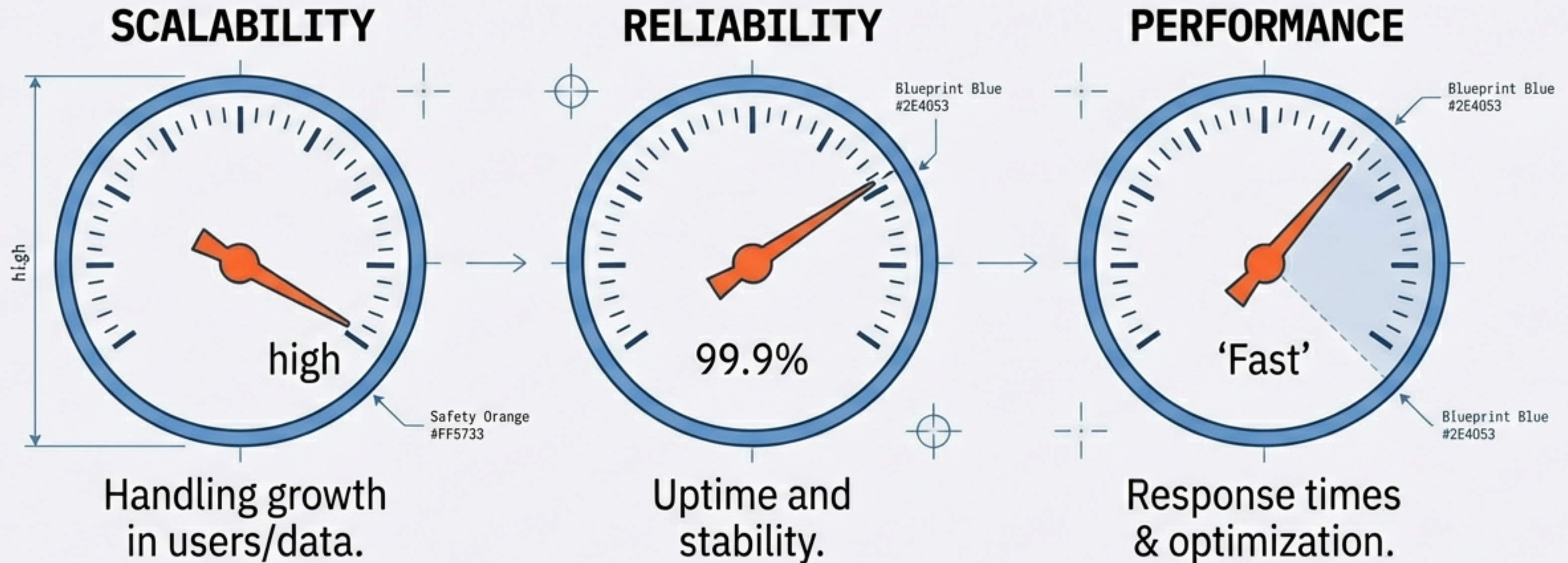


The Revert
A defined path to restore the previous state.

Preparing for the worst ensures the confidence to proceed.

NFRs: The Architecture of Sustainability

Defining HOW the system behaves, not just WHAT it does.



Non-Functional Requirements (NFRs) must be defined before development begins.

Operational Continuity & Continuous Improvement

MONITOR

Logs, Alerts,
Health Metrics

RESPOND

Disaster Recovery,
Capacity Plan

LEARN

Retrospectives,
Post-Mortems

IMPROVE

Optimize NFRs

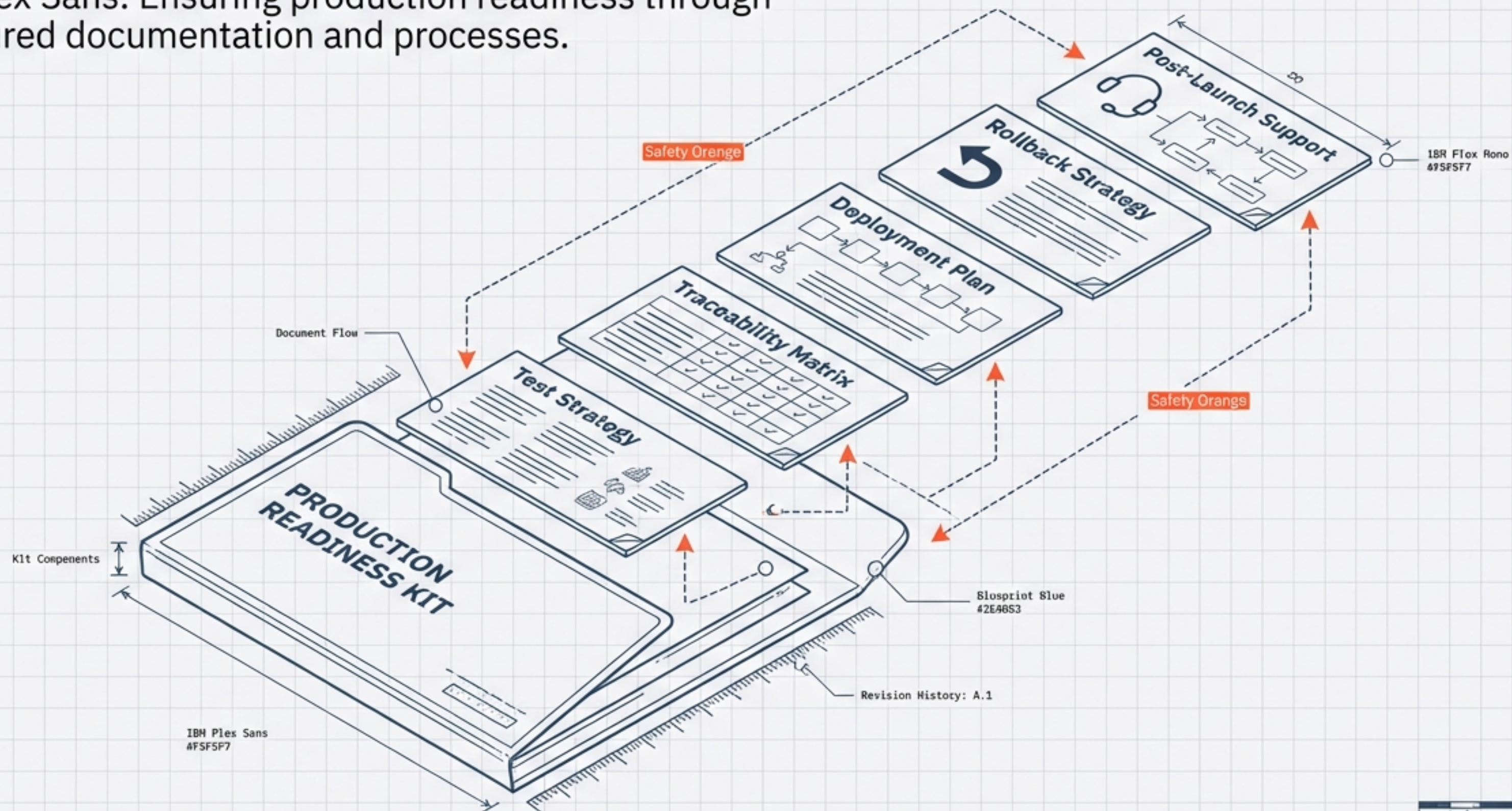


Consistent feedback ensures long-term operational resilience and performance optimization.

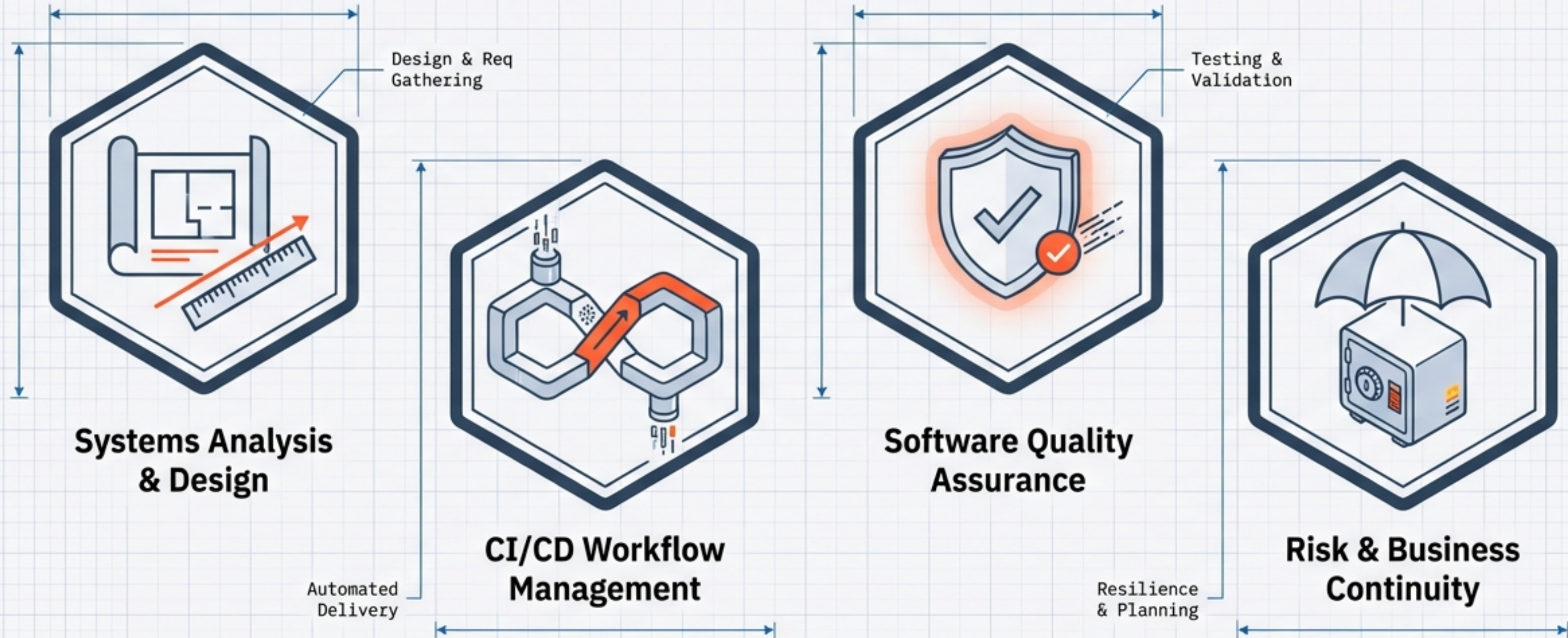
IBM Plex Sans (Drafting Black)

The Capstone: The Testing & Release Planning Kit

IBM Plex Sans: Ensuring production readiness through structured documentation and processes.



Core Competencies of the Systems Analyst



**You are not just delivering code.
You are engineering trust, stability, and business value.**